

RESEARCH ARTICLE | DECEMBER 01 1996

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Rev. Sci. Instrum. 67, 4057–4060 (1996)

<https://doi.org/10.1063/1.1147550>



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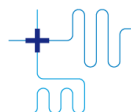
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Fabry–Perot interferometer in Brillouin scattering experiments

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(Received 20 March 1995; accepted for publication 23 August 1996)

The Fabry–Perot interferometer has become a standard spectroscopic tool in scientific laboratories for the study of Brillouin scattering. This article examines some basic properties of the Fabry–Perot interferometer with the student or beginning user in mind in order to demonstrate why it is especially useful for this application. Single and complex Fabry–Perot systems in use for Brillouin scattering studies are discussed starting from elementary equations to show, in particular, how the half-width of the transmission function depends upon the instrumental configuration. The primary results are summarized in graphical form. © 1996 American Institute of Physics.
 [S0034-6748(96)04211-6]

I. INTRODUCTION

The Fabry–Perot interferometer has become a basic tool for studies using the Brillouin scattering method. Studies using Brillouin spectroscopy accelerated rapidly in the early 1960s when laser light sources became routinely available.¹ The history of the Brillouin method shows several main stages of advancement connected to improvements in Fabry–Perot interferometers and methods for employing them.^{2–4} As a result of improving technique it became possible to study new materials, and this is traced in changes of scientific emphasis in the field.

The high intensity and small spectral width of a laser line allows the detection of Brillouin signals. However the Brillouin spectra for some kinds of materials contain lines which cannot be measured by the standard Fabry–Perot interferometer. The contrast and resolving power are too low to measure very weak Brillouin signals in the presence of the elastically scattered light. By use of the single and tandem multiple-passed interferometers a huge increase in contrast and resolution is obtained. This allows experiments in samples where the scattered light contains many observable spectral features. In this article we describe how the half-intensity width of the transmission function depends upon the instrumental configuration. We have taken this under consideration because it determines the resolving limit of the Fabry–Perot interferometer and consequently the field of applications.

II. THE SINGLE MULTIPLE-PASSED INTERFEROMETER

The problem for physicists is how to reproduce from the interference pattern the shape of Brillouin lines with a half-width of the order of 0.1 GHz or less. We ought to write out some basic definitions describing an interferometer and then find out what the essence of the problem is.

The Fabry–Perot interferometer is a tool in which multi-beam interference phenomenon between two flat or confocal high reflecting surfaces is applied. Following Born and Wolf⁵ the resultant difference of the phase δ between recombining wave fronts is equal to

$$\delta = 4\pi n \frac{d}{\lambda} \cos \beta + \phi = 2m\pi, \quad (1)$$

where n is the refractive index of the medium between reflecting surfaces, β is the angle of refraction for the ray passing through one of the mirrors, d is the distance between mirrors, λ is the light wavelength in a vacuum, ϕ is the additional change of the total phase in reflecting layers (especially in metallic layers), and m is the integer number. In further calculations changes in ϕ with respect to λ changes will be neglected. The characteristic interference pattern is a set of concentric rings focused on a screen. The pattern periodicity is a consequence of the periodicity of the transmission function which can be written as follows:

$$t(\delta) = \frac{I_E}{I_0} = \left(1 - \frac{A}{1-R}\right)^2 \left(\frac{1}{1+F \sin^2(\delta/2)}\right),$$

$$F = \frac{4R}{(1-R)^2}, \quad (2)$$

where R , T , and A are the reflection, transmission, and absorption coefficient of the reflecting layer, respectively and I_E and I_0 are the intensities of light on a screen and the intensity of light falling on the interferometer, respectively. The first quadratic in the power term describes stray loss caused by reflection and absorption in the reflecting layer. ($R+T+A=1$ results from the law of conservation of energy).

Two basic parameters can be determined to describe the characteristic of the Fabry–Perot interferometer transmission function. They are:

(a) The half-width of the transmission function ϵ_p .

This is a range of the resultant phase difference δ in the vicinity of maximum transmission where the transmission function has half of its maximum value. When we take into account symmetry of the $t(\delta)$ function we can write

$$\delta = 2m\pi \pm \frac{\epsilon_p}{2}, \quad 1 + F \sin^2\left(\frac{\epsilon_p}{4}\right) = 2. \quad (3)$$

Generally $F \gg 1$ and thus $\epsilon_p \ll 1$. Then

$$\epsilon_p = \frac{4}{\sqrt{F}}. \quad (4)$$

(b) Resolving power of the interferometer R_λ .

This quantity, defined as $R_\lambda = \lambda/\Delta\lambda$, indicates the ability of the instrument to separate wavelengths whose values are very close. Assuming that changes of the total phase δ which correspond to the resolved difference of two lines $\Delta\lambda_p$ are smaller than half-width ϵ_p of the interferometer we have

$$\epsilon_p = \delta(\lambda + \Delta\lambda) - \delta(\lambda) \approx 4\pi nd \frac{\Delta\lambda_p}{\lambda^2}, \quad (5)$$

and taking into account Eq. (4),

$$R_\lambda = \frac{\lambda}{\Delta\lambda} \approx \frac{\pi nd}{\lambda} \sqrt{F} (F \gg 1) \quad \text{or} \quad R_\lambda = \frac{\lambda}{\Delta\lambda} \approx \frac{4\pi nd}{\lambda \epsilon_p}, \quad (6)$$

which means that the resolving power of the interferometer is inversely proportional to the half-width of the transmission function. The procedure for reproducing from the interference pattern the shape of weak and narrow Brillouin lines is to increase the resolving power by decreasing the half-width of the transmission function. This is possible by use of the single and tandem multiple-passed interferometers.

For the single Fabry–Perot interferometer with number of passes $k > 1$ the transmission function can be written as follows:

$$t_k(\delta) = \left(1 - \frac{A}{1-R}\right)^{2k} \left(\frac{1}{1+F \sin^2(\delta/2)}\right)^k \quad (7)$$

or

$$t_k(\delta) = \frac{t_k(\delta)}{t_{k \max}(\delta)} = \left(\frac{1}{1+F \sin^2(\delta/2)}\right)^k, \quad (8)$$

where the factor responsible for losses was omitted (normalization). Taking into account the definition of the half-width, and the condition $\delta = 2m\pi \pm \epsilon_p^{(k)}/2$ which indicates that the transmission function is equal to 1/2, we obtain

$$\epsilon_p^{(k)} = \frac{4}{\sqrt{F}} (2^{1/k} - 1)^{1/2}, \quad \epsilon_p^{(1)} = \frac{4}{\sqrt{F}}. \quad (9)$$

Thus, the following conclusion is reached: a half-width of the multiple-passed single interferometer is smaller than $\epsilon_p^{(1)}$ —the half-width of the single interferometer. We can estimate a number of passages for practical applications. Taking into consideration $R=0.93$, $T=0.06$, and $A=0.01$ we have $F=759.2$. The calculations are presented in Table I. We can see that six passes are the upper limit if we take into account the intensity of light falling at the detector and a half-width as a criterion. This conclusion is not precise and in practice depends on the quality of a specific detection system and the investigated media. In liquids, glasses, and some crystals a single interferometer might be applied with the number of passes of light equal or greater than one. For instance Thomas *et al.*⁶ worked with a triple-passed system for single mode optical fibers or Dil and Brody⁷ used a five-passed arrangement for liquid metals.

TABLE I. The half-width changes and the light intensity decrease vs the number of the passages in a single Fabry–Perot interferometer.

k -number of passages	$\epsilon_p^{(k)}/\epsilon_p^{(1)}$	t_k/t_1
1	1.00	1.00
2	0.64	0.73
3	0.51	0.54
4	0.43	0.40
5	0.39	0.29
6	0.35	0.21
7	0.32	0.16
8	0.30	0.12
9	0.28	0.08

III. THE TANDEM MULTIPLE-PASSED INTERFEROMETER

The tandem arrangement is a combination of two Fabry–Perot interferometers placed one after another in a parallel manner (in the simplest case) with various distances d_1 and d_2 between mirrors. The distance between interferometers must be large enough to avoid distortion caused by reflections from the external surfaces of the mirrors.

The tandem system is used generally for nontransparent media where photons act on surface phonons and also on bulk phonons. The picture of the interaction is more complicated. Sapriel *et al.*⁸ and Takagi with Gammon⁹ used triple-passed systems for superlattices and thin Mylar and Pyrex layers, respectively. Five-passed systems were used by Polian *et al.*¹⁰ to study elastic properties of GaS under high pressure and by Kueny *et al.*¹¹ to watch magnon modes in superlattices. The six-passed system for determining elastic constants of a Mo/Ta superlattice was used by Bell *et al.*¹²

Construction of the tandem arrangement is not only a problem of mechanics but also a problem of aligning the multiple-passed system. Below we describe the theoretical condition for an appropriate adjustment.

The transmission function of the tandem arrangement is a product of transmission functions corresponding to the individual interferometers:

$$\vartheta(\delta_1, \delta_2) = \left(\frac{T_1^2}{(1-R_1)^2}\right) \left(\frac{T_2^2}{(1-R_2)^2}\right) \times \left(\frac{1}{1+F_1 \sin^2(\delta_1/2)}\right) \left(\frac{1}{1+F_2 \sin^2(\delta_2/2)}\right) \quad (10)$$

or

$$\vartheta'(\delta_1, \delta_2) = \frac{\vartheta(\delta_1, \delta_2)}{\vartheta_{\max}(\delta_1, \delta_2)} = \left(\frac{1}{1+F_1 \sin^2(\delta_1/2)}\right) \times \left(\frac{1}{1+F_2 \sin^2(\delta_2/2)}\right), \quad (11)$$

where $F_1 \neq F_2$ in general. As we can see it is a function of two variables δ_1 and δ_2 . The changes of δ_1 and δ_2 can be realized simultaneously or independently. In such a situation,

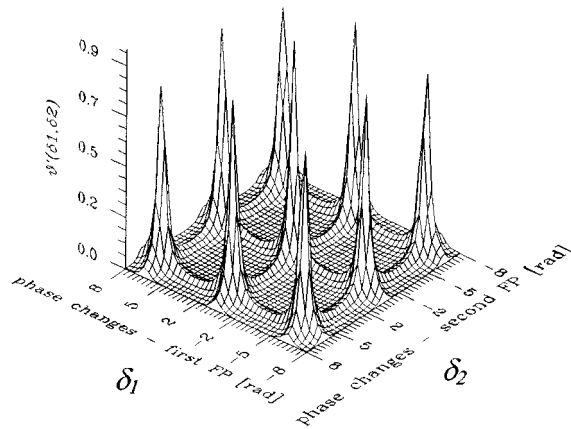


FIG. 1. Spatial plot of the $\vartheta'(\delta_1, \delta_2)$ function. δ_1 is the resultant difference of the phase in the first interferometer, δ_2 is the resultant difference of the phase in the second interferometer.

the transmission function can be examined on the (δ_1, δ_2) surface. The plots of $\vartheta'(\delta_1, \delta_2)$ are presented in Figs. 1 and 2.

As we can see from Fig. 1 if we want to obtain any interference image in the tandem setup two conditions must be achieved simultaneously and this is a basic condition of alignment,

$$\delta_1 = 4\pi \frac{n_1 d_1}{\lambda} \cos \beta_1 + \phi_1 = 2m_1\pi, \quad m_1 \in N, \quad (12)$$

$$\delta_2 = 4\pi \frac{n_2 d_2}{\lambda} \cos \beta_2 + \phi_2 = 2m_2\pi, \quad m_2 \in N.$$

Fulfilling the condition (12) for divergent flux of light assures the appearance of at least one ring in the interference

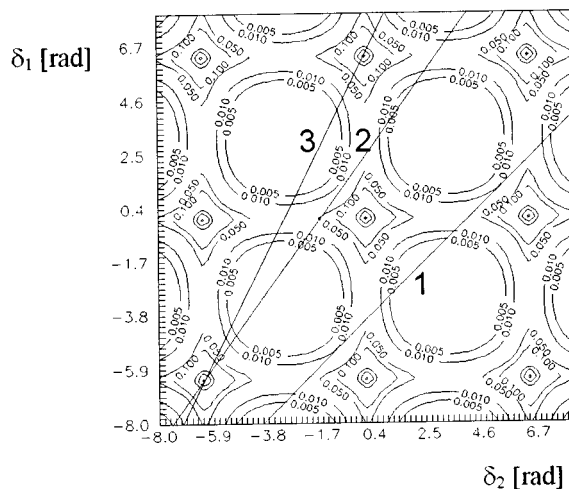


FIG. 2. Contour plot of the surface from Fig. 1. δ_1 is the resultant difference of the phase in the first interferometer, δ_2 is the resultant difference of the phase in the second interferometer. The straight lines correspond to linear dependence, Eq. (15): straight line 1 shows a lack of synchronization of the tandem, straight line 2 shows the partial synchronization of the tandem, straight line 3 shows the full synchronization of the tandem with $a=2$.

pattern (straight line 2 in Fig. 2). But in practice it is necessary to obtain more rings. Then additional conditions must be fulfilled (straight line 3 in Fig. 2)

$$\delta_1 = 2(m_1 - L_1)\pi, \quad \delta_2 = (2m_2 - L_2)\pi, \quad L_1, L_2 \in N. \quad (13)$$

We may establish that $L_2 \geq L_1$. The contrary condition describes an exchange of the interferometers but this does not influence the interference image. From Eq. (12) we have

$$a = \frac{L_2}{L_1} = \frac{d_2 n_2 \cos \beta_2}{d_1 n_1 \cos \beta_1} \geq 1, \quad a \in W. \quad (14)$$

The “ a ” value must be a rational number [see Eq. (13)] or very close to such number (generally not too big). The accuracy of the a value must be established in a such way as to avoid a situation when the shift of maximum values of the transmission functions is much greater than corresponding half-widths. In real situations adjustment can be made by changes in the refractive index or the distance between mirrors of one of the interferometers. However, the basic condition for the angles of incidence β_1 and β_2 is their identity. Moreover, both interferometers must lie on a common axis the same as the optical axis of the whole arrangement. The axis of the interferometers is normal to the surface of mirrors. These conditions assure that the interference pattern occurs as a set of concentric rings which are concentric and regular in intensity. In such a case, subsequent linear formula for the total phases of two interferometers becomes

$$\delta_2 = a\delta_1 + \phi_t, \quad \phi_t = \phi_2 - a\phi_1. \quad (15)$$

If we want to achieve the transmission function of the tandem setup we must intersect the plot from Fig. 1 with a plane perpendicular to the (δ_1, δ_2) surface. The direction of the cut is fixed by linear dependence, Eq. (15). The curve of the intersection can be projected on the (ϑ, δ_1) plane, and a typical picture of the transmission function of the tandem arrangement can be seen. The formula for this function can be written as

$$\vartheta_1(\delta_1) = \left(\frac{1}{1 + F_1 \sin^2(\delta_1/2)} \right) \times \left(\frac{1}{1 + F_2 \sin^2[(a\delta_1 + \phi_t)/2]} \right). \quad (16)$$

From the relation (15) arises

$$\phi_t = 2(m_2 - am_1)\pi \quad (17)$$

and we can conclude that the period of the transmission function given by Eq. (16) is equal to $2L_1\pi$; and $L_1 + L_2 - 1$ weak additional orders are visible between main orders.

The basic problem for the tandem is to determine a half-width of the main order of the transmission function. It is obvious that for a general case [Eq. (10)] in which the peaks are not necessarily aligned this parameter has no physical sense. But for the relation (16) we can apply the same criterion as for the single interferometer. For the function given by Eq. (16) we have

$$\delta_1 = 2(m_1 - L_1)\pi \pm \epsilon_{t1}/2, \quad (18)$$

where ϵ_{t1} is determined from the condition

$$[1 + F_1 \sin^2(\epsilon_{t1}/4)][1 + F_2 \sin^2(a\epsilon_{t1}/4)] = 2, \quad (19)$$

and assuming that $\epsilon_{t1} \ll 1$ we have

$$\epsilon_{t1} = \frac{4\sqrt{2}}{[(F_1^2 + 6F_1F_2a^2 + F_2^2a^4)^{1/2} + F_1 + F_2a^2]^{1/2}}. \quad (20)$$

Moreover, from Eq. (15) we can write $\epsilon_{t2} = a\epsilon_{t1}$. It is easy to observe that the following conditions are always achieved:

$$\epsilon_{t1} < \epsilon_{p1}, \quad \epsilon_{t2} < \epsilon_{p2}, \quad (21)$$

which means that a half-width of the tandem is always smaller than half-widths of the single interferometer.

Note that because the free spectral range is determined by the interferometer having a smaller distance between mirrors, and that the values of d_1 and d_2 must be chosen in such a way as to avoid overlapping the Brillouin peaks with the main orders m_1 and $(m_1 - L_1)$ and with the weak orders $(m_2 - 1), (m_2 - 2), \dots, (m_2 - L_2 - L_1 + 2)$ from the interferometer having the larger distance between mirrors.

Making similar considerations for the tandem interferometer with the number of passages $k > 1$ we have

$$\epsilon_{t1}^{(k)} = \frac{4\sqrt{2}(2^{1/k} - 1)^{1/2}}{\{[F_1^2 + F_2^2a^4 + 2F_1F_2a^2(2^{1+1/k} - 1)]^{1/2} + F_1 + F_2a^2\}^{1/2}}. \quad (22)$$

By substitution to the above formula $k=1$ we obtain Eq. (20) and by the $F_1=0$ substitution we get the formula (9) valid for the single multiple-passed interferometer. Determination of these parameters enables us to define the resolving power of the tandem arrangement by use of the simplest criterion [see Eq. (5)],

$$R_{t\lambda} = \frac{4\pi n d_1}{\lambda \epsilon_{t1}^{(k)}}, \quad (23)$$

where we have taken $\epsilon_{t1}^{(k)}$ from Eq. (22). In the above formula an assumption that $n_1 = n_2 = n$ was taken. This means that both interferometers are placed in the same medium (for example in the same pressure chamber).

We have shown the transmission function dependence upon the instrumental configuration for single and complex Fabry–Perot systems in use for Brillouin scattering studies. The half-width of this function decreases with the number of passes through the interferometers. Then the resolving power of the system increases. This allows for experiments whereby many features of the Brillouin spectrum are to be measured.

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